DR. MARIA BEATRIZ WALTER COSTA

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SUMMARY

Bioinformatician with **PhD** and **10+ years**' experience in **analysis of large-scale biological datasets**. Proven track record in generating biologically meaningful insights to **support data-driven decisions**. Published author (*BMC Bioinformatics, JMIR Medical Informatics*) with advanced expertise in **Python, R and statistical modeling**.

WORK EXPERIENCE

Friedrich-Schiller University Jena — Jena, Germany
Research Associate (fixed-term) — *10/2022 - 09/2025*

- Led project on large-scale microbial omics data to identify environmental biomarkers (**ML, Python**)
- Designed **scalable high-throughput pipeline** for genomics data (extracts 150k+ features from 13k+ genomes), leveraging parallel computing on high-performance clusters (Snakemake)
- Co-supervised a doctoral researcher and contributed to teaching in large-scale data analysis

University of Leipzig Medical Center — Leipzig, Germany
Research Associate (fixed-term) — *02/2020 - 12/2021*

- Applied **statistical methods** to clinical data (260k+ patients, 20M+ laboratory values, 4 publications) to **support decision** in automated diagnosis of 3 acute conditions (Statistics, R)
- Worked closely with physicians and industry partners to build AMPEL Decision Support System
- Structured a clinical knowledge dataflow to improve downstream analysis (SAP-based data)

Brazilian Agricultural Research Corporation (EMBRAPA) — Brasília/DF, Brazil
Consultant (fixed-term) — *11/2018 - 03/2019*

- Analyzed differential expression data to find key genes to optimize ethanol production (R, DESeq2)
- Collaborated with experimental scientists to support decision-making in strain engineering

University of Leipzig — Leipzig, Germany
Research Associate (fixed-term) — *04/2017 - 03/2018*

- Developed a **new statistical method**, `SSS-test` (*BMC Bioinformatics*), and discovered with it 14 ncRNA biomarkers with potential relevance to psychiatric disorders (**Bioinformatics**)

EDUCATION

Dr. rer. nat. in Computer Science (*magna cum laude*) — Leipzig, Germany
University of Leipzig, Chair of Bioinformatics — *2013 - 2018*

- Specialization in method development in Computational Biology (6 publications, 3 methods)

M. Sc. in Molecular Biology Brasília/DF, Brazil
University of Brasília, Graduation Program of Molecular Biology *2010 - 2012*

· Specialization in NGS-based analysis in immunogenomics (1 publication)

B. Sc. in Biological Sciences Brasília/DF, Brazil
University of Brasília, Institute of Biological Sciences *2006 - 2010*

· Specialization in Bioinformatics

LANGUAGES

English: near-native speaker (C2, Cambridge CPE)
Portuguese: Native speaker (C2)
German: Advanced (C1)
Spanish: Advanced (B2/C1)

TECHNICAL SKILLS

Programming: Python, R, Perl, Bash, SQL (mySQL), AI-assisted coding (Gemini)
Bioinformatics: genomics, transcriptomics, AI/ML: machine learning, statistical modeling
Workflow: Snakemake, Git (GitHub)
Data Processing: Linux/Unix, HPC platforms (Slurm, SGE)

SELECTED PRODUCTS

- `FxTractor`: reproducible high-throughput pipeline for feature extraction from microbial genomes (150k+ features across 13k+ genomes), leveraging parallel computing on HPC platforms

  `https://github.com/MGXlab/FxTractor`

- `SSS-test`: the first tool of the scientific community that detects adaptive selection in ncRNAs, demonstrated by a publication in *BMC Bioinformatics*

  `https://github.com/waltercostamb/SSS-test`

SELECTED PUBLICATIONS

- **Walter Costa MB** *et al.* The Clinical Decision Support System AMPEL for Laboratory Diagnostics: Implementation and Technical Evaluation. *JMIR Med Informatics*, 2021

  `https://medinform.jmir.org/2021/6/e20407`

- **Walter Costa, MB** *et al.* SSS-Test: A novel test for detecting selection on the secondary structures of non-coding RNAs. *BMC Bioinformatics*, 2019

  `https://doi.org/10.1186/s12859-019-2711-y`